

CHARLES BENELLO

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EDUCATION

The University of Chicago

Chicago, IL

Master of Science in Financial Mathematics

Expected Mar 2027

- Wayne Booth Graduate Student Teaching Prize (2026) - UChicago's top graduate teaching award
- Coursework: Distributed Systems, ML in Software Engineering, Agentic AI, Reinforcement Learning

The University of Chicago | GPA: 3.75/4.0

Chicago, IL

Bachelor of Science in Mathematics and Computer Science (Systems), Minor in Korean

Jun 2025

- Coursework: Database Systems, Operating Systems, Algorithms, Concurrency Control, Computer Architecture, Networks

EXPERIENCE

Incoming Software Development Engineer Intern

Aug – Nov 2026

Amazon Web Services (Redshift)

Palo Alto, CA

- Joining the Amazon Redshift query optimization team to work on plan-time decisions and cost models for analytical workloads.

Database Research Assistant

Apr 2023 – Present

ChiData Research Lab, University of Chicago

Chicago, IL

- First-authored VLDB 2026 paper (CrocSort) by designing a parallel external merge sort in Rust that completes under tight memory budgets where PostgreSQL, DuckDB, and ClickHouse abort, jointly planning memory and phase-specific thread counts.
- Co-authored CIDR 2025 paper on resource-adaptive query execution with paged memory management, adapting operators to runtime contention.
- Sole-authored HPTS 2026 paper on a lightweight prediction model for sort configuration on cloud and disaggregated hardware, transferring calibration across machines within 3–5% of optimal without brute-force sweeps.
- Improved random access speed by 200% and reduced memory footprint by 40% by engineering Small String Optimization across storage structures, inlining short keys directly into nodes to eliminate heap allocations and pointer chases on the hot path.

GPU Systems Researcher

Jun – Sep 2025

EPFL (DIAS Lab, Prof. Anastasia Ailamaki)

Lausanne, Switzerland

- Engineered a macroblock-based GPU decompression strategy in CUDA/C++ that cut out-of-memory query times by 50–200%.
- Built cost-based simulation models enabling adaptive decompression strategy selection at plan time on TPC-DS workloads.
- Designed join patterns and resource contention models across I/O, decompression, and compute, shaping a new cost-based research direction.

Software Engineering Intern

Feb 2026 – Present

Alter Domus

Chicago, IL

- Lifted document extraction accuracy from 50% to 92–97% via a multi-stage pipeline with schema-enforced output for 100+ page financial filings.
- Built an LLM-as-a-Judge evaluation framework benchmarking faithfulness, numerical accuracy, and completeness across document types.

Quantitative Researcher

Mar 2026 – Present

PayPal (UChicago Project Lab)

Chicago, IL

- Pretrained a TabBERT transaction foundation model on 74M MBD transactions (Sberbank, 200K users) and fine-tuned cross-dataset to IBM TabFormer; partial fine-tuning lifted FM AUC from 0.74 to 0.996, with stacked XGBoost+FM saving \$215K per 2K-user test cohort.
- Audited the original 16-feature XGBoost baseline (AUC 0.66) and built a 21-feature replacement reaching AUC 0.99; built 14 fusion variants (stacking, isotonic calibration, MLP) and per-user diagnostic dashboards (t-SNE, kNN, KMeans).

PROJECTS

GPU External Sort | CUDA, C++ | github.com/cmhenello/gpu-research

2025–Present

- Built a GPU external merge sort in CUDA (port of CrocSort, VLDB 2026). On TPC-H SF50 ORDER BY lineitem (300M rows, 36 GB), beats DuckDB by 20.1×, Polars by 34.5×, and MendSort (JouleSort 2023) by 2.4× on a PCIe-Gen3 GPU. Cost-based plan-time selector between compact-key and full-key paths cuts PCIe amplification to 0.14× via key-only upload, permutation gather, and triple-buffered streaming.

Concurrent Database Engine | Rust | github.com/cmhenello/CrustyDB

2025

- Built a multi-threaded relational database from scratch in Rust: B+ tree indexing, concurrent buffer pool with page-level locking, heap storage, and a SQL query execution engine.

Transit Coverage Optimizer | Python | cmhenello.github.io/covtsp

2026

- Beat the Guinness World Record by 60 minutes (16h45m vs. 17h45m) with an automated Covering TSP pipeline on a time-expanded graph (160K nodes, 1.26M edges) using k-NN with urgency scoring and epsilon-greedy search. Generalized to NYC (475 stations) and Berlin.

TECHNICAL SKILLS

Languages: C/C++, Python, Rust, CUDA, Go, SQL, TypeScript, JavaScript, Bash

Tools: PostgreSQL, DuckDB, Databricks, Docker, Git, Linux, PyTorch, FastAPI, Next.js, SLURM

Concepts: Distributed Systems, Database Internals, Query Execution & Optimization, Concurrency, GPU Programming, Performance Engineering, Cost-Based Optimization, System Design

PUBLICATIONS

- [1] R. Otaki*, C. Benello*, et al. “CrocSort: Resource-Efficient, Skew-Resilient Parallel External Merge Sort.” *VLDB 2026*. (*Equal contribution)
- [2] C. Benello. “Getting the Most Out of External Sorting in Cloud and Disaggregated Environments.” *HPTS 2026*.
- [3] R. Otaki, J.H. Chang, C. Benello, A. Elmore, G. Graefe. “Resource-Adaptive Query Execution with Paged Memory Management.” *CIDR 2025*.

ADDITIONAL

Languages: Korean, Italian (Intermediate); French (Basic) | **Interests:** Running (Chicago Marathon '22, '23, '24), Climbing, Weightlifting